



Physics-Informed Extreme Learning Machine framework for solving linear elasticity mechanics problems

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ABSTRACT

In neural network-based methods for elasticity mechanics such as Physics-Informed Neural Networks (PINNs), the training process is typically time-consuming due to multiple optimization iterations and reliance on automatic differentiation for gradient computation. This paper proposes a novel framework that combines Physics-Informed Extreme Learning Machines (PIELM) with linear elastic mechanics, focusing on the integration of discretization operators with the principles of linear elasticity mechanics. These discretization operators are constructed based on the form of higher-order derivatives in PIELM, along with governing equations and boundary conditions. Physical information is incorporated into the loss function through discretization operators applied at collocation points. The solution of the proposed method is calculated through the least squares method, which determines the output weights. The comparison with PINN under nonlinear loads and geometric defects demonstrates that PIELM achieves superior error control and prediction accuracy. Furthermore, PIELM achieves more precise in evaluated error norms compared to Finite Element Method (FEM) in a three-dimensional problem with an analytical solution. In conclusion, the proposed method offers an robust and efficient alternative method for linear elasticity.

1. Introduction

In the modern fields of engineering and physical sciences, problems pertaining to linear elasticity mechanics occupy a pivotal position (Barber, 2002). Such problems are of great importance in the fields of structural engineering (Vesic, 1961), materials science (Heitkam et al., 2016) and mechanical design (Arieli et al., 2024). These applications not only reflect the theoretical importance of linear elastic mechanics, but also emphasize the necessity for efficient and accurate solution methods. Despite their widespread use in the past decades, traditional numerical methods, such as the finite element method (FEM) (Hrennikoff, 2021; Liu et al., 2022), finite difference method (FDM) (Liu, 2018; Morton and Mayers, 2005), and finite volume method (FVM) (Eymard et al., 2000; LeVeque, 2002), have inherent limitations in dealing with complex geometries, high computational cost, numerical stability, and accuracy (Haghighat et al., 2021; Tvergaard, 2016).

In recent years, there has been a growing application of neural network-based methods, particularly Physics-informed Neural Networks (PINNs) (Karniadakis et al., 2021), to the field of computational mechanics. Currently, the principal methodologies for solving PDEs in computational mechanics are the governing equation based method

and the energy based method. Samaniego et al. (2020) proposed a Deep Energy Method (DEM) and applied to computational mechanics for the first time, where the system's energy is employed as the neural network's loss function to solve mechanical problems, offering a new and efficient approach. Nguyen-Thanh et al. (2020) explored the application of the DEM for modeling finite deformation in hyperelastic materials, showcasing its effectiveness in capturing complex material behaviors. Nguyen-Thanh et al. (2021) further proposed the Parametric DEM (PDEM), which defines the loss function based on the total potential energy of the system and minimizes it in the reference domain, and is suitable for elasticity problems considering strain gradient effects. Guo et al. (2019) proposed a deep collocation method (DCM) for analyzing the bending of Kirchhoff plates, the DCM is designed to minimize a loss function derived from governing equations and boundary conditions. Haghighat et al. (2021) proposed a PINN framework designed for inversion problems and surrogate modeling in elasticity mechanics, incorporating the PDE term and boundary conditions into the loss term. Based on DCM, Huang and Peng (2024) combined coordinate transformation and finite difference approximation to propose the Deep Difference Collocation Method (DDCM), which calculates derivatives

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through convolutional kernels, making it suitable for irregular domains. Gu et al. (2023) presented a new simulation framework based on PINNs for modeling crack problems, which is trained on discretely sampled collocation points and incorporates crack-tip asymptotic functions. Fuhg and Bouklas (2022) proposed the mixed DEM (mDEM), integrating strong form, energy form, and constitutive equations in the loss function to optimize equilibrium residuals and material consistency, improving accuracy for traction boundary conditions and stress concentration. Li et al. (2021) compared two approaches governing equations based and energy based for implementing PINN in modeling the behavior of elastic plates, evaluating their respective strengths and weaknesses. These methods leverage the capabilities of deep learning and are capable of handling intricate geometries and boundary conditions without the need for traditional meshing. However, despite these advances, PINNs still present significant challenges in terms of prolonged training times, difficulties in selecting optimal network sizes, and limitations in solution accuracy and efficiency.

The Extreme Learning Machine (ELM) (Huang et al., 2006), a rapid single-layer feed-forward neural network training method, has garnered interest due to its stochastic initialization of hidden layer weights and non-iterative optimization process. ELM has been effectively employed for the expeditious processing of extensive datasets and the resolution of a multitude of physical problems. The integration of physical information into ELM model has led to the development of the Physics-informed Extreme Learning Machine (PIELM) (Dwivedi and Srinivasan, 2020), a novel approach that significantly enhances the efficiency and accuracy of solutions by incorporating physical constraints directly into the loss function. Dong and Yang (2022) developed an algorithm to optimize the hyperparameters of PIELM for computational PDEs, resulting in enhanced solution performance when compared to both classical and high-order finite element methods. Li et al. (2024) introduced a coupled ELM (CELM) method incorporated with physical laws to solve PDEs problem with high accuracy, while comparing different activation functions influence on performance of CELM. Dong and Li (2021) proposed a local ELM (locELM) method for solving linear and nonlinear PDEs by dividing the domain into sub-domains and optimizing the output layer through linear or nonlinear least squares. Wang and Dong (2024) expanded the ELM method to encompass high-dimensional PDEs and introduced a novel approach, termed the Approximate Theory of Functional Connections (A-TFC), which approximates the TFC expression by retaining key terms, thereby enhancing efficiency and facilitating the handling of boundary conditions in high-dimensional settings. Liu et al. (2023) employed the Bayesian PIELM (BPIELM) to address forward and inverse PDEs problems, facilitating precise prediction and uncertainty quantification in the presence of noise while reducing the computational burden. Although PIELM has been shown to be highly precise and efficient in a number of different applications, its potential in the field of elasticity mechanics remains to be seen. Therefore, the PIELM framework for solving elasticity mechanics merits further exploration, as it offers both accuracy and efficiency solution.

The objective of this paper is to develop a PIELM framework for solving linear elasticity problems, with the aim of providing a streamlined and high-precision for solution process. This article is structured as follows: Firstly, a comprehensive theoretical background is presented, including the linear elasticity equation and the PIELM method. Secondly, the framework of PIELM for linear elasticity mechanics problems is introduced, describing the neural network framework, the construction of discretization operators, the formulation of the loss function and the training process. Thirdly, two two-dimensional plates with nonlinear loads and geometric defects, along with a three-dimensional self-weight elongation problem with analytical solution, are evaluated to demonstrate its efficacy. Finally, the influence of parameters on the performance of this framework as well as its differences from PINN across various aspects are discussed.

2. Theoretical foundation

This section introduces the fundamentals of linear elasticity mechanics and the process of PIELM solving PDEs. It provides the theoretical support for combining PIELM and linear elasticity mechanics in subsequent sections.

2.1. Basics of linear elasticity mechanics

In the context of linear elasticity problems, the solution to the problem can be represented by a set of governing equations and boundary conditions.

In the elastomer domain(Ω), the displacement field u satisfies the following PDEs(Equilibrium equation) :

$$A\sigma + \bar{f} = 0 \quad (1)$$

where A is a differential operator, $\bar{f}$ is volume load vector and σ is stress vector. Under small deformation, stress and strain conform to geometric equations:

$$\sigma = D\epsilon \quad (2)$$

where D is elastics matrix, which is made of Lamé constant(G and λ) and ϵ is strain vector, which can be calculated by geometric equation:

$$\epsilon = Lu \quad (3)$$

where $L = A^T$ is a differential operator.

In the elastic boundary($\partial\Omega$), the governing equations presented above must satisfy the following boundary conditions:

$$\begin{cases} u = \bar{u} \\ n\sigma = \bar{T} \end{cases} \quad (4)$$

where $\bar{u}$ and $\bar{T}$ are determined by the displacement constraints and external load condition in determined boundary and n is the direction vector matrix of coordinate points.

2.2. physics-informed extreme learning machine

The Extreme Learning Machine (ELM) is a feed-forward neural network learning algorithm, which was proposed by Huang et al. (2006). Its core idea is to randomly initialize the weights and biases from the input layer to the hidden layer during the learning process, and then directly compute the weights from the hidden layer to the output layer by parsing method. This approach enables ELM to achieve a notable enhancement in training speed relative to traditional neural network algorithms, while maintaining a satisfactory degree of generalization ability. The framework of ELM with N input nodes and M output nodes is illustrated in Fig. 1.

The input layer consists of N input nodes, forming the input vector $x = [x_1, x_2, \dots, x_N]_{1 \times N}$. Each input node is responsible for receiving one feature of the input data. The hidden layer contains L neurons, each employing an activation function $g(\cdot)$. The transformation from the input layer to the hidden layer is characterized by randomly generated the weight matrix $w = [w_1^T, w_2^T, \dots, w_D^T]_{D \times L}$ and the bias vector $b = [b_1, b_2, \dots, b_L]_{1 \times L}$. Every input node is connected to each hidden neuron, forming a fully connected architecture. The output of each neuron h_i in hidden layer is computed using the activation function applied to the weighted sum of the inputs plus the bias:

$$h_i = g_i(xw_i^T + b_i) \quad (5)$$

where g is activation function. The choice of activation function can significantly impact the smoothness and accuracy of ELM. Common activation functions include the sigmoid function, hyperbolic tangent (tanh), radial basis functions (RBF) and sin function.

The output layer comprises M nodes, forming the output vector $u = [u_1, u_2, \dots, u_M]_{1 \times M}$. Each hidden neuron is connected to every

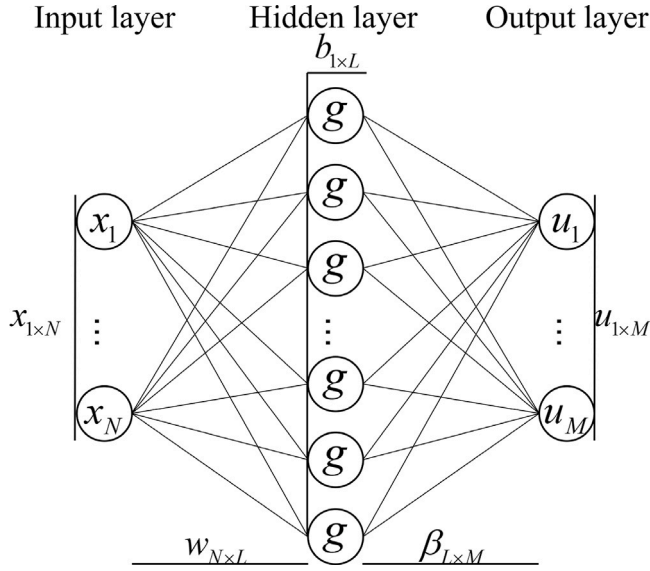


Fig. 1. Framework of ELM.

output node through the weight matrix $\beta = [\beta_1, \beta_2, \dots, \beta_M]_{L \times M}$. The final output u_j for each output node is calculated by summing the contributions from all hidden neurons:

$$u_j = h\beta_j \quad (6)$$

where $h = [h_1, h_2, \dots, h_L]_{1 \times L}$ is hidden neurons vector.

Formally, this entire process can be represented as Eq. (7). This equation encapsulates the ELM's operation, demonstrating how input data is progressively transformed through the network to generate the desired outputs.

$$u = g(xw + b)\beta \quad (7)$$

The traditional ELM is well-suited for solving regression and classification problems due to its advantages in terms of accuracy and efficiency. To further enhance the capabilities of ELM in applications involving physical systems and scientific computing, the concept of Physics-Informed Extreme Learning Machines (PIELM) (Dwivedi and Srinivasan, 2020) was introduced. PIELM integrates domain-specific physical laws and constraints into the ELM framework, ensuring that model predictions are not only data-consistent but also adhere to underlying physical laws. This approach results in more robust and accurate models for complex problems.

Combining PDEs with ELM leverages the rapid training capabilities of ELM while embedding physical constraints directly into the learning process. Consider the following PDE:

$$\begin{cases} L(u(x)) = f(x) & x \in \Omega \\ B(u(x)) = h(x) & x \in \partial\Omega \end{cases} \quad (8)$$

where L is a differential operator and B is a boundary operator. In the context of PDEs, differential operators play a pivotal role in elucidating the behavior of the system, the higher derivatives of the output with respect to the input in L expressing the PDE in terms of ELM can be expressed as:

$$\frac{\partial^n u}{\partial x_i^n} = w_i^n \circ g^{(n)}(xw + b)\beta \quad (9)$$

where $\circ$ is element-wise multiplication. For linear differential operators, it can be expressed as:

$$f(x) = L(u(x)) = L(g(xw + b))\beta, x \in \Omega \quad (10)$$

Boundary conditions are crucial in PDEs. In practice, boundary conditions a uniquely determined solution for the ELM framework. The boundary equation can be expressed in ELM:

$$h(x) = B(g(xw + b))\beta, x \in \partial\Omega \quad (11)$$

Thus the residual term can be expressed as:

$$\xi = \left\| \begin{bmatrix} f(x_i) \\ h(x_j) \end{bmatrix} - \begin{bmatrix} L(g(x_iw + b)) \\ B(g(x_jw + b)) \end{bmatrix} \beta \right\|_2^2, x_i \in \Omega, x_j \in \partial\Omega \quad (12)$$

where β is the weight vector which need to be solved. Let $K = [f(x_i), h(x_j)]^T$, and $H = [L(g(x_iw + b)), B(g(x_jw + b))]^T$, β can be solved using the least squares method ensures that the solution adheres to both the differential equation and the boundary conditions:

$$\beta = \text{pinv}(H) \cdot K \quad (13)$$

where $\text{pinv}(H)$ is generalized inverse of H .

3. Method

This section outlines the methodology employed to solve linear elasticity problems using the PIELM. It discusses the neural network framework, the construction of discretization operators, the formulation of the loss function and the training process. Each step is optimized for the specific requirements of linear elasticity. The resulting framework adheres to the governing physical laws while maintaining computational efficiency.

3.1. Network framework for linear elasticity mechanics problems

In the PIELM used for solving linear elasticity mechanics problems, the framework is specifically designed to map spatial coordinates to displacement fields, which are crucial for understanding the material deformation under applied loads. The framework consists of an input layer, hidden layers and an output layer, and each component is tailored to efficiently handle the computational requirements of elasticity problems.

The input and output of the ELM are designed to reflect the physical nature of the problem. The input layer takes spatial coordinates as inputs. The input vector includes coordinates (x, y, z) for a 3D problem and (x, y) for a 2D problem. These inputs represent the positions within the material where the displacement needs to be calculated.

The output layer produces the displacement components corresponding to the input coordinates. The rationale for choosing displacements as outputs is that both stress and strain can be derived from the displacement field in ELM framework using the governing equations and the constitutive relationships. The output vector includes displacement components (u, v, w) for a 3D problem and (u, v) for a 2D problem. The output dimensions match the input dimensions, reflecting the displacements at the specified coordinates.

The choice of activation function in hidden layer is essential for the network's ability to learn and generalize. In the context of linear elasticity problem, it is necessary to compute second-order derivatives as shown in Eq. (1). In order to better fit the range of the hidden layer, the $\tanh$ function has been chosen as the activation function. This function exhibits a bounded range in its higher-order derivatives, which is less susceptible to changes in the input coordinates than other activation functions.

In summary, the PIELM framework for a 3D problem can be constructed as follows,

$$\begin{aligned} \text{Input layer: } x &= [x, y, z] \in \mathbb{R}^3 \\ \text{Hidden layer: } h &= \tanh(xw + b) \in \mathbb{R}^L \\ \text{Output layer: } u &= [u, v, w] = h\beta \in \mathbb{R}^3 \end{aligned} \quad (14)$$

the input and output layers can be replaced with $x = [x, y] \in \mathbb{R}^2$ and $u = [u, v] = h\beta \in \mathbb{R}^2$ to get PIELM framework for a 2D problem.

3.2. Discretization operators

The governing linear elasticity PDEs need be discretized in the PIELM framework in order to enable the network to approximate the solution in an efficient manner. This involves converting the continuous PDEs into a form that can be handled by the PIELM, with the equations then evaluated using collocation points. The key idea underlying this process is to express both the PDEs and their derivatives in terms of the parameters and activation functions of the aforementioned PIELM.

The governing equations of linear elasticity involve the balance of loads and the constitutive relationships between stress and strain. These equations can be expressed in terms of the displacement field components u , v , and w . The network approximates these displacement components as outputs, which are functions of the spatial coordinates. The divergence of the stress tensor in terms of the displacement components can be expressed as:

$$\mathbf{ADL}u + \bar{\mathbf{f}} = 0 \quad (15)$$

in which, the elements of $\mathbf{ADL}$ can be represented as

$$\mathbf{ADL}_{ij} = (\lambda + G) \frac{\partial^2}{\partial x(i)\partial x(j)} + \delta_{ij} G \sum_{k=1}^3 \frac{\partial^2}{\partial x(k)^2} \quad (16)$$

where δ_{ij} is defined as:

$$\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \quad (17)$$

and the 2-order partial derivative can be calculated by Eq. (9):

$$\frac{\partial^2 u(i)}{\partial x(j)\partial x(k)} = w_j \circ w_k \circ \tanh''(\mathbf{x}\mathbf{w} + \mathbf{b})\beta_i \quad (18)$$

therefore, Eq. (15) can be rewritten as:

$$\mathbf{K}\beta' + \bar{\mathbf{f}} = 0 \quad (19)$$

in which, $\beta' = [\beta_1, \beta_2, \dots, \beta_M]^T$, $\mathbf{K}$ is the governing discretization differential operator and the element is obtained by replacing the 2-order partial derivative $\frac{\partial^2}{\partial x(i)\partial x(j)}$ of $\mathbf{ADL}$ with $w_j \circ w_k \circ \tanh''(\mathbf{x}\mathbf{w} + \mathbf{b})$.

Similarly, the boundary conditions can be expressed in terms of displacements:

$$\begin{cases} \mathbf{P}\beta' = \bar{\mathbf{u}} \\ \mathbf{Q}\beta' = \bar{\mathbf{T}} \end{cases} \quad (20)$$

where $\mathbf{P}$ is the displacement discretization operator, the element can be expressed as:

$$P_{li} = \mathbf{0} \text{ or } \mathbf{h} \quad (21)$$

in which, $\mathbf{h}$ is the vector of output from hidden layer, the value of the element is related to the actual displacement constraint. $\mathbf{Q}$ is the stress discretization differential operator, the element can be expressed as:

$$Q_{ij} = \lambda n_i w_j \circ \tanh'(\mathbf{x}\mathbf{w} + \mathbf{b}) + G n_j w_i \circ \tanh'(\mathbf{x}\mathbf{w} + \mathbf{b}) + \delta_{ij} G \sum_{k=1}^3 n_k w_k \circ \tanh'(\mathbf{x}\mathbf{w} + \mathbf{b}) \quad (22)$$

In this framework, the displacement components serve as the primary variables, from which the stress and strain can be derived. The PIELM framework employs a discretization operator at each collocation point, enabling the enforcement of the governing equations and boundary conditions with great efficiency. This approach ensures that the PDEs and boundary conditions in terms of displacements are inherently satisfied within the model, leading to accurate and efficient solutions.

3.3. Construction of the loss function

In the PIELM framework, the construction of the loss function represents a pivotal stage in ensuring that the network accurately

approximates the solution to the linear elasticity problem. The loss function of PIELM consists of several parts, each corresponding to a different aspect of the problem. The incorporation of the governing PDEs and boundary conditions into the loss function facilitates the embedding of the physics of the problem within the optimization process.

The primary constituent of the loss function is the residual of the governing PDEs, evaluated at the collocation points. The residual at each collocation point x_i can be defined as follows:

$$\mathcal{L}_{PDE} = \sum_{x_i \in \Omega} \|\mathbf{K}_i \beta' + \bar{\mathbf{f}}_i\|^2 \quad (23)$$

where Ω denotes the set of collocation points in intradomain, $\mathbf{K}_i$ represents the governing discretization differential operator applied at the collocation point x_i , and $\bar{\mathbf{f}}_i$ is the body force vector at x_i .

The loss function incorporates terms for both dirichlet and Neumann boundary conditions. The residual for the dirichlet boundary conditions is:

$$\mathcal{L}_{BC,u} = \sum_{x_j \in \partial\Omega_u} \|\mathbf{P}_j \beta' - \bar{\mathbf{u}}_j\|^2 \quad (24)$$

where $\partial\Omega_u$ denotes the set of boundary collocation points where displacements are prescribed, $\mathbf{P}_j$ is the displacement discretization operator at the collocation point x_j , and $\bar{\mathbf{u}}_j$ is the prescribed displacement vector at x_j .

Similarly, the residual for the Neumann boundary conditions is:

$$\mathcal{L}_{BC,T} = \sum_{x_k \in \partial\Omega_T} \|\mathbf{Q}_k \beta' - \bar{\mathbf{T}}_k\|^2 \quad (25)$$

where $\partial\Omega_T$ denotes the set of boundary collocation points where tractions are prescribed, $\mathbf{Q}_k$ is the stress discretization differential operator at the collocation point x_k , and $\bar{\mathbf{T}}_k$ is the prescribed traction vector at x_k .

The total loss function $\mathcal{L}$ for the PIELM is a weighted sum of these components:

$$\mathcal{L} = \mathcal{L}_{PDE} + \alpha_u \mathcal{L}_{BC,u} + \alpha_T \mathcal{L}_{BC,T} \quad (26)$$

where α_u and α_T are weights that balance the contributions of the boundary condition residuals relative to the PDE residuals. It can be assembled into a single linear system of equations:

$$\begin{pmatrix} \mathbf{K} \\ \alpha_u \mathbf{P} \\ \alpha_T \mathbf{Q} \end{pmatrix} \beta' = \begin{pmatrix} -\bar{\mathbf{f}} \\ \alpha_u \bar{\mathbf{u}} \\ \alpha_T \bar{\mathbf{T}} \end{pmatrix} \quad (27)$$

This transformation results in the formation of a classical PIELM loss term, which can then be solved using the formula provided in Eq. (13).

3.4. Overall workflow

The training process of the PIELM for solving linear elasticity mechanics problems comprises the generation of training data, the initialization of the network, the construction of loss function based on the discretization operators at collocation points, and the solution of the output weights. This process ensures that the network accurately represents the solution to the linear elasticity problems. Fig. 2 provides a visual representation of the key components and flow of this process.

The key components and flow of the process in Fig. 2 are detailed as follows:

1. Collocation Points Generation

- Intra-domain collocation points
- Dirichlet boundary collocation points
- Neumann boundary collocation points

2. Network Initialization

- Randomly generate parameters weights $\mathbf{w}$ and bias $\mathbf{b}$ for input layer.

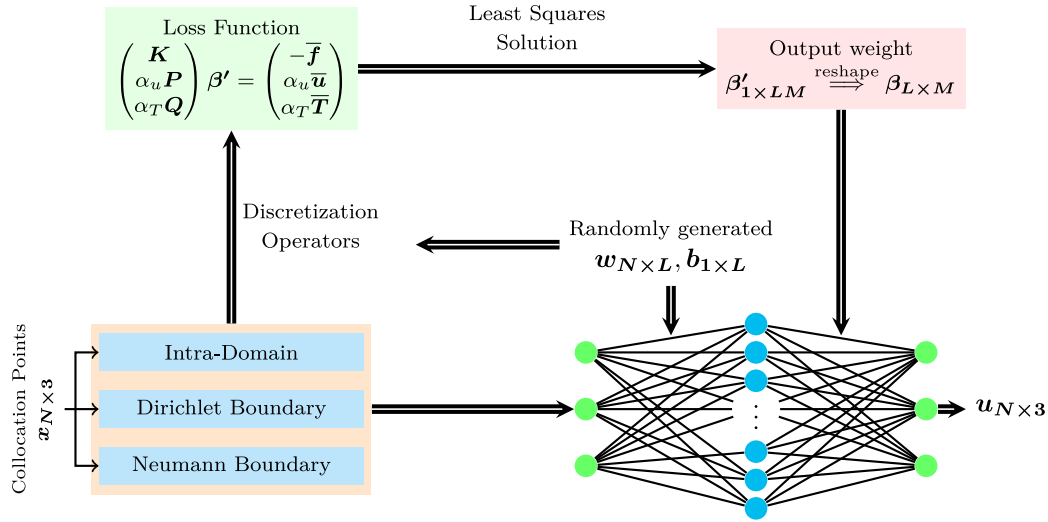


Fig. 2. Overall workflow of PIELM for solving linear elasticity mechanics problems.

3. Loss Function Construction

- Construct discretization operators to transform continuous problems into a computable discrete format.
- Use matrix representation to formalize the loss function.

4. Calculate output weights

- Solve for the output weights using the least squares method.
- Reshape output weights to fit output layers.

5. Network Output

- Integrate weights across input and output layers to compute the final network output.
- Calculate and present the displacement output, the result of network computations.

In summary, this training procedure delineates the overarching process of PIELM, which guarantees that the network accurately approximates the solution of linear elasticity mechanics by generating the training data, initializing the network, calculating the error based on physical constraints and optimizing the output weights.

4. Case studies

In order to demonstrate the efficacy of the proposed framework in addressing problems within the field of linear elasticity mechanics, two plate cases which are characterized by nonlinear loads and geometric defects conditions and another three-dimensional case with analytical solution will be presented. This will enable a comprehensive understanding of the PIELM's capabilities with regard to the management of different boundary conditions and external loads. A comparison will be drawn between the predictive outputs' accuracy and those obtained through FEM and PINN methodologies.

4.1. Case1: Plate with non-linear distributed load

Focusing on the effect of nonlinear loads, the presented problem concerns a $2 \times 2(m \times m)$ square plate subjected to a distributed load along its vertical edges (Fig. 3). The left and right sides are subjected to a nonlinear distributed load, which is defined by the equation $q = y^2$ (Pa), and upper and lower sides are load-free. The Lamé constant of material

is $G = 30$ Pa and $\lambda = 20$ Pa. The coordinates are taken as the point of origin and the original model (Fig. 3(a)) is thus simplified into a quarter model (Fig. 3(b)) due to the symmetry of the load and the geometry. Equivalent boundary conditions for the quarter model have been annotated in Fig. 3(b).

A series of collocation points distributed throughout the domain, including boundaries, must first be identified. The sets of collocation points, designated as x_1 , x_2 , x_3 , x_4 , and x_5 , respectively, are positioned as follows: x_1 is within the domain, x_2 is on the left side, x_3 is on the below side, x_4 is on the upper side, and x_5 is on the right side. These points should then be transformed into a loss function through the utilization of appropriate discretization operators. The loss function can be constructed as $H\beta' = K$, in which, $K = [0, 0, 0, 0, 0, \alpha_T x_2^2(2), 0]^T$ is constituted by known boundary conditions and H is constituted by discretization operators:

$$H = \begin{bmatrix} K(x_1) \\ \alpha_u P_1(x_2) \\ \alpha_u P_2(x_3) \\ \alpha_T Q_1(x_4) \\ \alpha_T Q_2(x_5) \end{bmatrix} \quad (28)$$

in which, $x = [x, y]$ is the vector of coordinate. The displacement discretization operators in loss function are as follows:

$$P_1(x_2) = [h(x_2), 0], P_2(x_3) = [0, h(x_2)] \quad (29)$$

where, the discrepancy is determined by the directions of displacement constraints. The stress discretization differential operator in loss function are as follows:

$$\begin{aligned} Q_1(x_4) &= \begin{bmatrix} Gw_2 & \lambda Gw_1 \\ \lambda w_1 & (\lambda + 2G)w_2 \end{bmatrix} \otimes \tanh'(x_4 w + b) \\ Q_2(x_5) &= \begin{bmatrix} (\lambda + 2G)w_1 & \lambda w_2 \\ Gw_2 & Gw_1 \end{bmatrix} \otimes \tanh'(x_5 w + b) \end{aligned} \quad (30)$$

A comparative experimental investigation is conducted on PIELM and PINN, with an identical number of collocation points and neurons. The FEM results are employed as a reference solution, with the node coordinates serving as inputs to ensure a consistent basis for comparison across all methods. The predicted u displacement solutions and errors compared to FEM are presented in Fig. 4. It is observed that both PIELM and PINN closely replicate the FEM solution. The error regions, as shown in Fig. 4(d)(e), reveal that PIELM tends to have smaller localized error zones, while PINN shows more dispersed error regions, revealing that PIELM performs better across the entire domain. The superiority of PIELM in prediction is further substantiated by the Mean Squared Error (MSE) values provided in the Table 1.

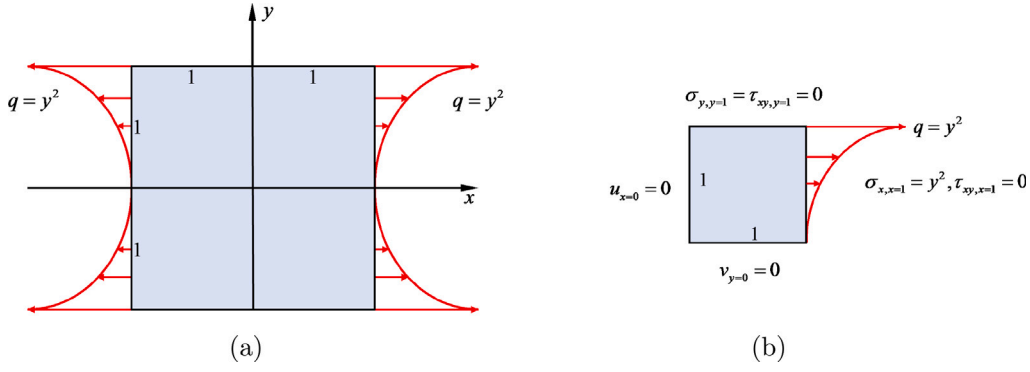


Fig. 3. Problem setup for plate with non-linear distributed load (a) Original model (b) One quarter model.

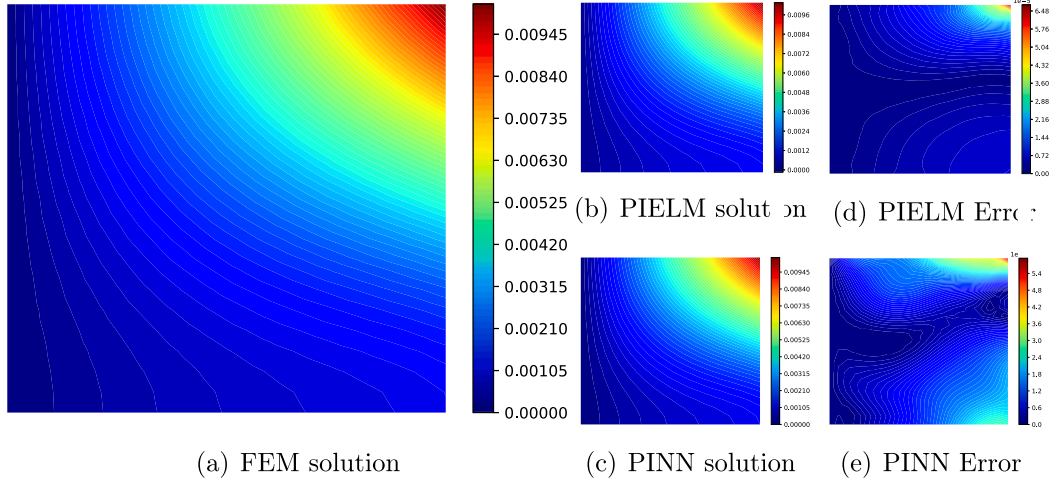


Fig. 4. Predicted case 1 u displacement solutions for FEM, PIELM and PINN: (a) FEM solution, (b) PIELM solution, (c) PINN solution, (d) PIELM Error compared to FEM, (e) PINN Error compared to FEM.

Table 1

Mean squared error of the case1 predicted displacement and stress.

Method	Mean Squared Error(MSE)					
	u	v	$\sqrt{u^2 + v^2}$	σ_x	σ_y	τ_{xy}
PIELM	5.72E-11	2.85E-11	7.20E-11	6.48E-06	3.10E-06	2.78E-05
PINN	1.28E-10	1.10E-10	1.65E-10	8.71E-06	9.80E-06	2.75E-05

The maximum error boundary, as illustrated in Fig. 5, represents the upper and right boundaries. The displacement u compared to FEM along both boundaries is presented in Fig. 5. PIELM performs well across the majority of the domain, however, its error tends to increase near top right corner and the maximum error of PIELM is greater than that of PINN. This is particularly evident in regions where displacement gradients are significant variations.

4.2. Case2: Plate with non-linear geometric defect

Focusing on the effect of nonlinear geometric defect, the second presented problem concerns a $2 \times 2(m \times m)$ square plate with a $r = 0.2$ circle defect subjected to a unified load along its vertical edges (Fig. 6). The left and right sides are subjected to a unified distributed load, which is defined by the equation $q = 1(Pa)$, and upper and lower sides are load-free. The Lamé constant of material is $G = 26.92$ Pa and $\lambda = 23.08$ Pa. Similarly, the quarter model is used for analysis (Fig. 6(b)).

The sets of collocation points, designated as x_1, x_2, x_3, x_4, x_5 and x_6 . The positions of collocation points $x_1 - x_5$ is same as case 1, x_6 is

on the circle edge. The loss function can be constructed as $H\beta' = K$, in which, $K = [0, 0, 0, 0, 0, 0, \alpha_T 1, 0, 0, 0]^T$ and H is:

$$H = \begin{bmatrix} K(x_1) \\ \alpha_u P_1(x_2) \\ \alpha_u P_2(x_3) \\ \alpha_T Q_1(x_4) \\ \alpha_T Q_2(x_5) \\ \alpha_T Q_3(x_6) \end{bmatrix} \quad (31)$$

in which, the displacement discretization operators P_1, P_2 and the stress discretization differential operator Q_1, Q_2 are same as Eqs. (29) and (30). The stress discretization differential operator Q_3 is as follow:

$$Q_3(x_6) = \begin{bmatrix} n_x(\lambda + 2G)w_1 + n_y Gw_2 & n_x \lambda w_2 + n_y Gw_1 \\ n_x \lambda w_1 + n_y Gw_2 & n_x(\lambda + 2G)w_2 + n_y Gw_1 \end{bmatrix} \quad (32)$$

in which, $n_x = -x_6(1)/r$ and $n_y = -x_6(2)/r$ are the direction cosine of the normal outside the boundary.

The predicted displacement and stress solutions are presented in Fig. 7, and the errors compared to FEM are presented in Fig. 8. The predictive capabilities of both the PIELM and PINN methodologies are precise in addressing nonlinear geometric defects in plates. Each method has been demonstrated to effectively capture the physical information of the concentrated stress. It is noteworthy that PIELM also demonstrates superior overall accuracy compared to the PINN in this case, as evidenced by its lower MSE values (Table 2) for various displacement and stress components.

The presence of defects in a plate results in the formation of stress concentrations around the defects, with stress values in these regions

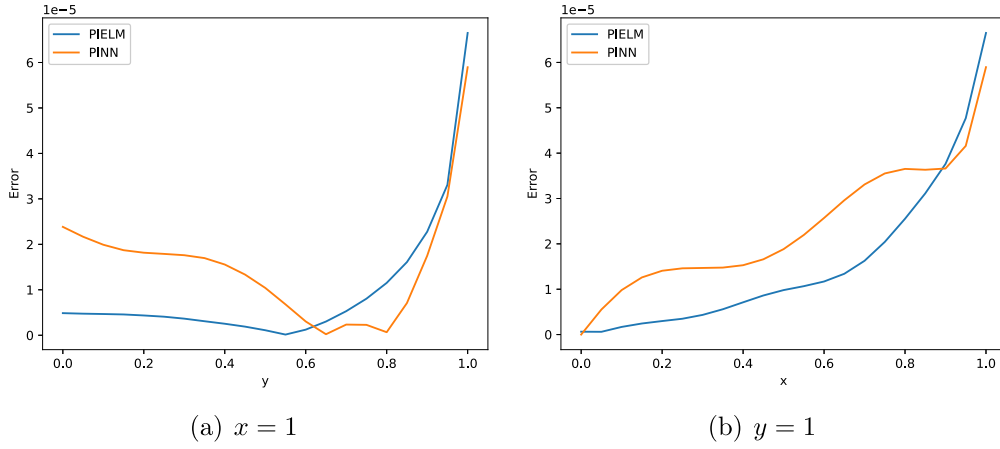


Fig. 5. Predicted case 1 displacement u error compared to FEM along $x = 1$ and $y = 1$: (a) $x = 1$, (b) $y = 1$.

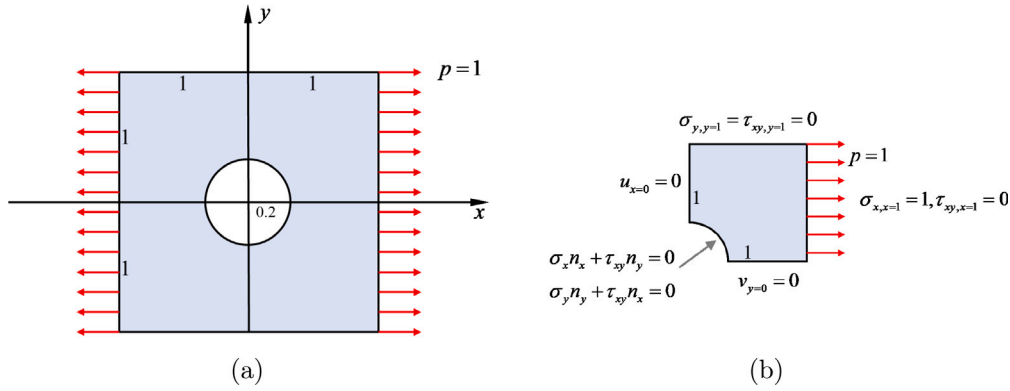


Fig. 6. Problem setup for plate with non-Linear geometric defect: (a) Original model (b) One quarter model.

Table 2

Mean squared error of the case 2 predicted displacement and stress.

Method	Mean Squared Error(MSE)					
	u	v	$\sqrt{u^2 + v^2}$	σ_x	σ_y	τ_{xy}
PIELM	6.28E-9	5.89E-9	1.02E-8	3.06E-4	3.54E-4	6.81E-5
PINN	3.74E-8	4.36E-8	6.46E-8	3.42E-4	8.86E-4	8.52E-5

Table 3

Error norms comparison of the Case 3.

Method	L^2	H^1 semi	Energy
PIELM	2.22E-1	8.75E-2	3.16
FEM	2.21E-1	2.34E-1	5.51

being significantly higher than in other parts of the plate. This leads to a steep stress gradient around the defect, which in turn makes the prediction of these regions more challenging. As illustrated in Fig. 8, the errors between the PIELM and PINN stress estimates are predominantly concentrated in the geometric defects. The error distributions of PIELM and PINN models on geometric defect are compared in Fig. 9. The results demonstrate that in the border region, the error of PIELM markedly increases and even surpasses that of the PINN model. This result indicates that although PIELM demonstrates proficiency in comprehensive error control, it exhibits limitations in its capacity to regulate errors when confronted with a steep gradient region. This discrepancy indicates that the PIELM model may require additional refinement to enhance its accuracy and stability in this region.

4.3. Case3: Self-weight elongation problem with analytical solution

A classical problem of elongation under self-weight is addressed in this section, with the analytical solution utilized as a benchmark for comparative analysis. As shown in Fig. 10, the differential element in entire cube is subjected to a gravitational force ρg , where ρ is the density and g is the gravitational acceleration. The parameters for

analysis is as follows: the length of cube is selected as $l = 1$, the density of material is selected as $\rho = 1$, the elasticity modulus is selected as $E = 200$ Pa and the Poisson ratio is selected as $\mu = 0.2$. The displacements analytical solution of this model is presented as follow:

$$\begin{cases} u = -\frac{\mu \rho g x z}{E} \\ v = -\frac{\mu \rho g y z}{E} \\ w = \frac{\rho g (z^2 + \mu(x^2 + y^2) - l^2)}{2E} \end{cases} \quad (33)$$

The FEM method is chosen for the comparison and the result of displacement solutions are illustrated in Fig. 11. It can be observed that both two method can approximate the analytical solutions well. To ensure the accuracy of the solution is rigorously assessed, multiple error norm including L^2 , H^1 -semi and energy norm are employed. Respectively, the L^2 norm is employed to quantify the overall displacement error, the H^1 semi-norm is utilized to assess gradient error, and the energy norm is utilized to assess system energy error. These norm errors provide a comprehensive evaluation of accuracy. A comparison of error norms for PIELM and FEM is provided in Table 3. The results indicate that PIELM achieve lower values in both H^1 semi and energy norm compared to FEM, providing a more accurate representation of gradient and energy characteristics. In terms of the L^2 norm, both the

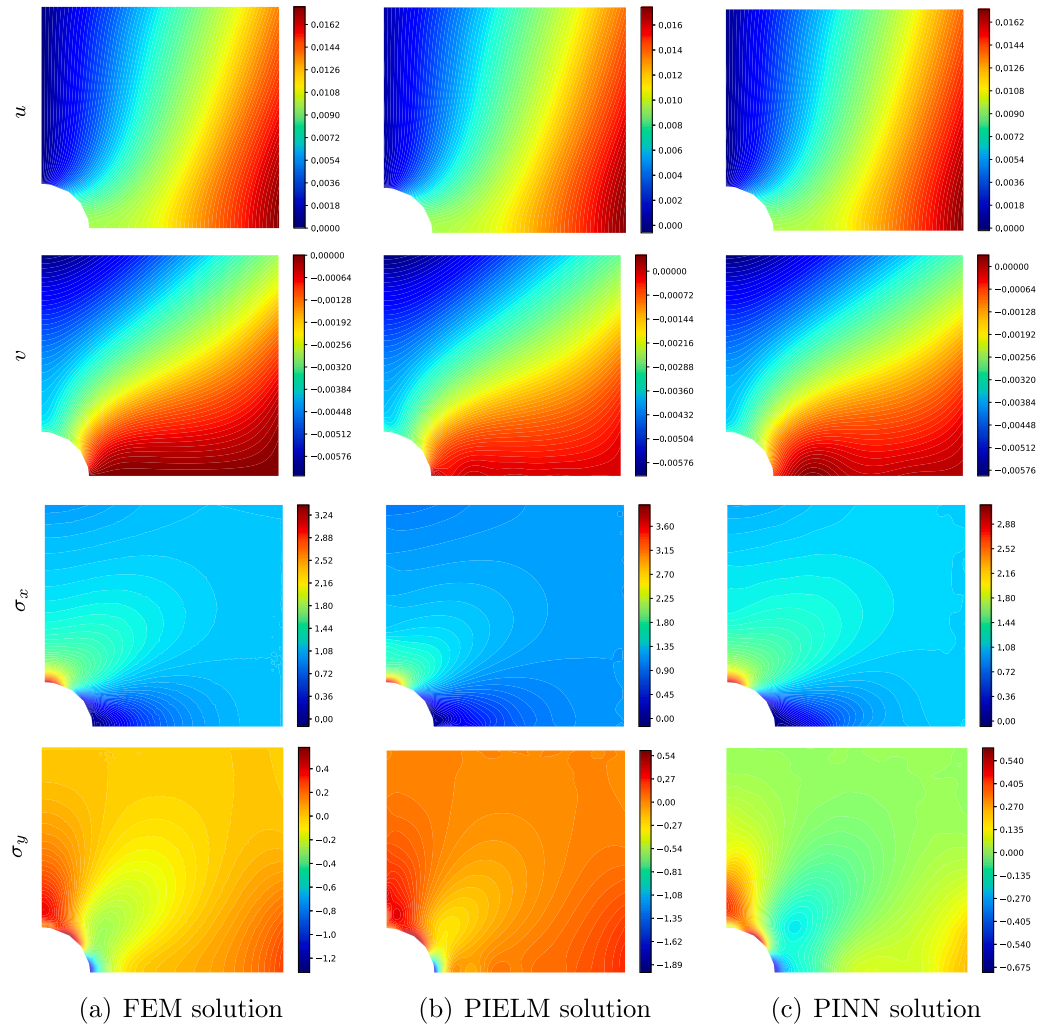


Fig. 7. Predicted case 2 displacement for FEM, PIELM and PINN: (a) FEM solution, (b) PIELM solution, (c) PINN solution.

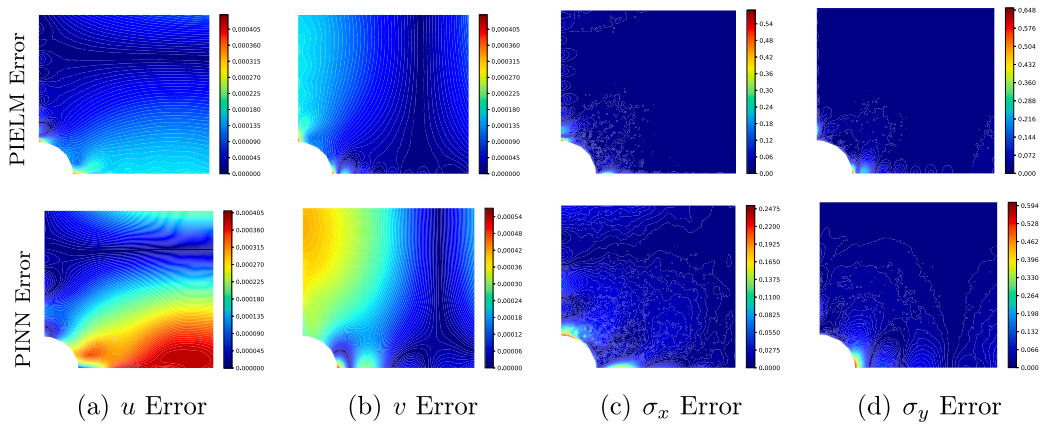


Fig. 8. Predicted case 2 error compared to FEM for PIELM and PINN: (a) u Error, (b) v Error, (c) σ_x Error, (d) σ_y Error.

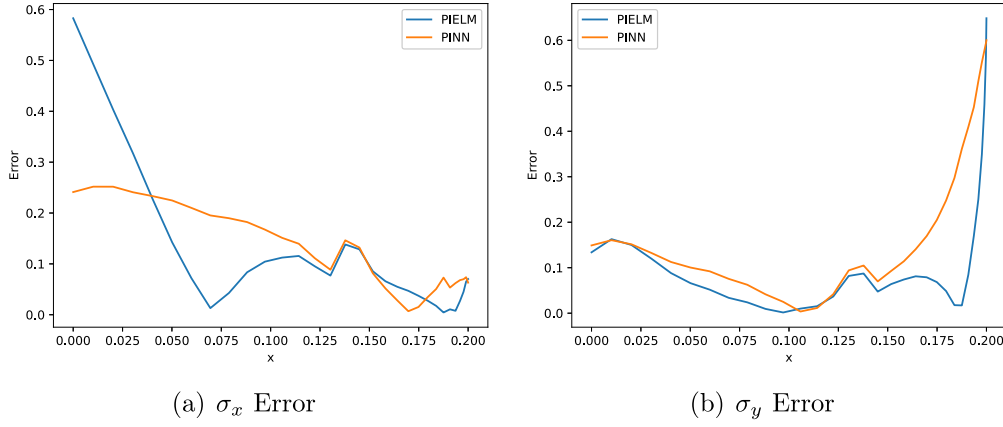


Fig. 9. Predicted case 2 stress error compared to FEM along geometric defect: (a) σ_x Error, (b) σ_y Error.

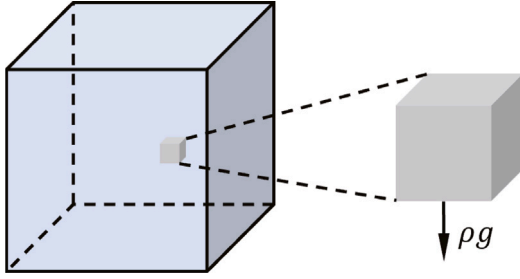


Fig. 10. Elongation model under self-weight.

methods perform similar result. In conclusion, the results demonstrate that PIELM is more precise in reproducing the model's behavior across the evaluated error norms in this case.

5. Discussion

5.1. Influence of the number of hidden layer neurons on PIELM solution

In the modeling process of the PIELM, the number of hidden layer neurons represents a crucial hyperparameter, exerting a direct impact on the expressiveness, convergence and computational efficiency of the model. In order to verify the influence of the number of hidden layers on the solution performance of PIELM, different numbers of layers are selected for analysis in Case 1 and Case 2. The MSE values under different layers in PIELM are presented in Fig. 12.

A reduction in the number of case 1 and case 2 neurons in the hidden layer has been observed to result in a notable increase in the MSE value, particularly with regard to the stress component. Conversely, as the number of layers has been increased, the MSE has been observed to decline, eventually reaching a point of stability. This suggests that in both cases, the model's prediction error has decreased significantly with an increased number of hidden layer neurons. However, a slower decline has been observed in case 2 compared to case 1, and it has been observed to stabilize at a level of approximately 2000. It is therefore necessary to increase the number of hidden layer neurons in order to enhance the predictive accuracy of the model when dealing with complex structures or stress concentration.

5.2. Influence of randomly initialization on stability of PIELM solution

Given that the weights in the PIELM are generated through random initialization, this randomness may introduce uncertainty in the model performance, particularly with regard to the accuracy of the solutions and the control of errors. To demonstrate this, Case 2 is selected for

analysis and repeated trials are conducted with input weights randomly generated from a standard normal distribution. The stability of PIELM is demonstrated by examining its MSE to FEM method across multiple trials. The MSE for the displacement components and the stress components are shown in Fig. 13.

The results for the displacement component demonstrate consistent performance with minimal fluctuations. Even in the presence of some outliers, the values remain within the same order of magnitude, thereby substantiating the reliability of PIELM displacement predictions. Similarly, the stress component maintained a low and stable MSE value over the trials. Overall, these findings confirm that performance of PIELM are not dependent on the random initialization of parameters.

5.3. Influence of local refined sampling on significant gradient variations region

In regions exhibiting significant gradient variations, the distribution of data exerts a direct influence on the model's capacity to discern the distinctive features of these regions. In the event of an insufficient number of data points in regions of critical importance, the model may encounter difficulties in accurately learning the complex behavior exhibited in these regions, which could potentially result in an increased prevalence of prediction errors. To further examine the influence of data distribution on local accuracy, an experiment was devised to incrementally augment the number of collocation points in regions exhibiting significant gradient variations, with the objective of discerning the tendency of maximum stress error. Case 2 is selected as the research object, and the density of collocation points in the vicinity of the geometric defect is increased in a stepwise manner, from 1000 to 9000 points, with an increment of 1000 points at each stage. As shown in Fig. 14, uniform sampling is used throughout the region (Fig. 14(a)) and local refined sampling is used at the radius of 0.2–0.3 (Fig. 14(b)).

At each point density, PIELM is used for training, and the error value at the point of maximum stress error predicted by the model is recorded, the result is presented in Fig. 15. As the number of sampling points increases, the model's accuracy in predicting stress in regions with significant gradient changes gradually improves. Once the number of sampling points reaches a certain threshold, the model's ability to capture complex local features significantly enhances, leading to a noticeable decrease in error. This indicates that to improve the model's predictive accuracy in regions with significant gradient variations, it is necessary to increase the number of sampling points in a reasonable manner.

5.4. Comparison between the PIELM and PINN framework

Architecture and Network Structure: The structures of PINN and PIELM are absolutely different. PINN employs Deep Neural Networks

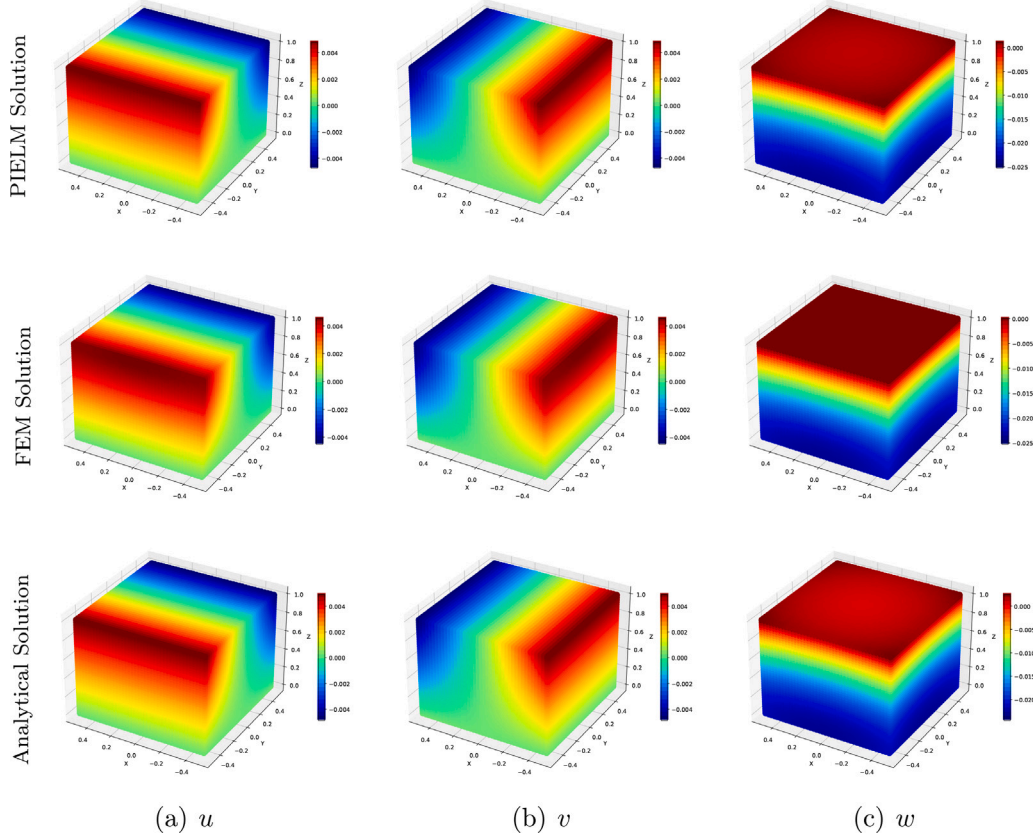


Fig. 11. Predicted case 3 displacement solutions for PIELM, FEM and Analytical solution: (a) u , (b) v , (c) w .

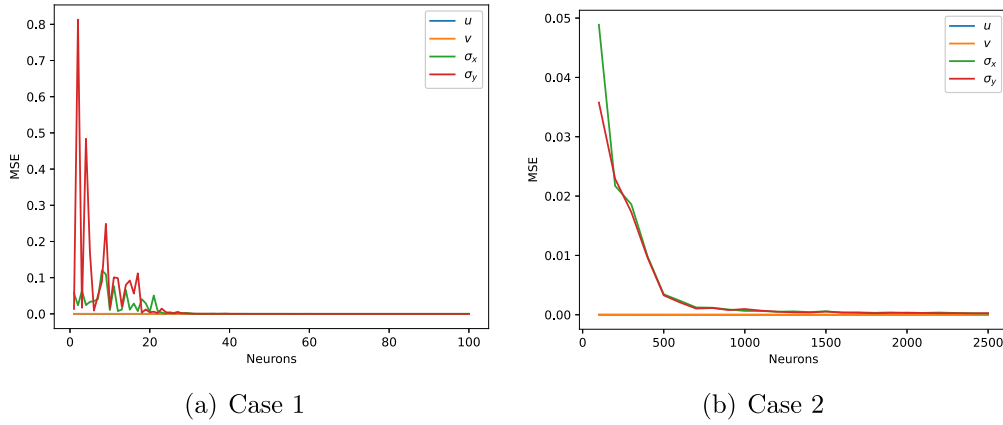


Fig. 12. MSE values under different layers in PIELM: (a) Case 1 (b) Case 2.

(DNNs) as its fundamental structural component. Typically, PINNs comprise multiple layers, each comprising a substantial number of neurons with intricate activation functions and weights. PIELM is based on ELM framework, which typically uses a single-layer feed-forward neural network. PIELM has a single hidden layer, with a variable number of neurons in hidden layer, and the activation function may be a simple non-linear function.

Hyperparameters: There is a notable distinction between PINN and PIELM with regard to the hyperparameters involved. PINN encompasses a multitude of hyperparameters, including the network structure (e.g., the number of layers and nodes), optimization parameters (e.g., the learning rate and the batch size), and so forth. The selection of these parameters has a significant impact on the model performance and is challenging to adjust. In contrast, PIELM has a smaller number of hyperparameters, with the primary focus being on the selection of

the number of nodes in the hidden layer and the activation function. These are relatively straightforward to adjust and have a lesser impact on the model performance.

Gradient Calculation: The principal distinction between PINN and PIELM in the context of gradient calculation pertains to the methodology and computational complexity. PINN employs the backpropagation algorithm for iterative optimization. The calculation process is complex and time-consuming, requiring the calculation and update of weights layer by layer to approximate the optimal solution. Furthermore, there is a possibility that PINN may encounter issues such as gradient explosion or gradient disappearance. PIELM, on the other hand, calculates the gradient directly through the derivative formula of the activation function by randomly initializing the weights. The gradient calculation is simple and fast, and does not involve iterative optimization.

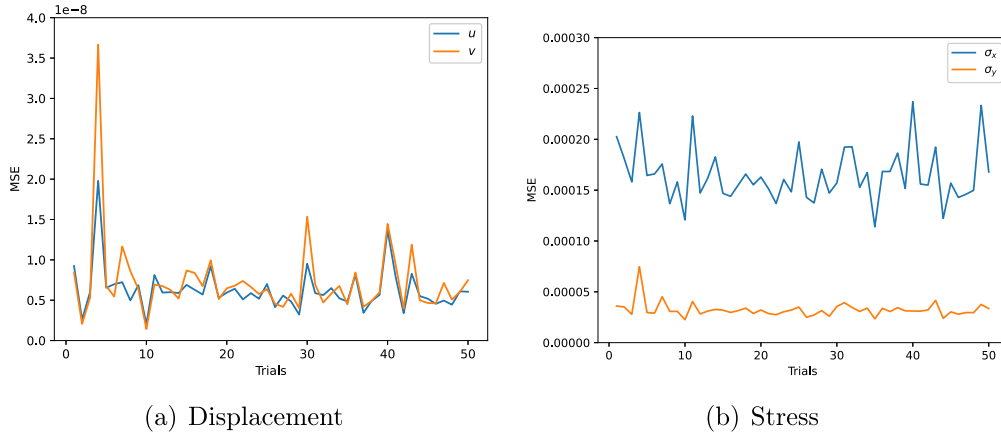


Fig. 13. Solution stability of PIELM across multiple trials:(a) Displacement, (b) Stress.

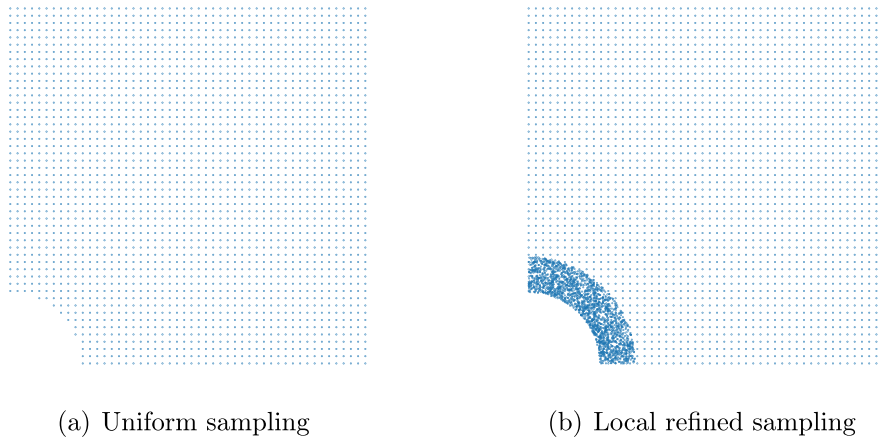


Fig. 14. Local refined sampling for Case 2: (a) Uniform sampling (b) Local refined sampling.

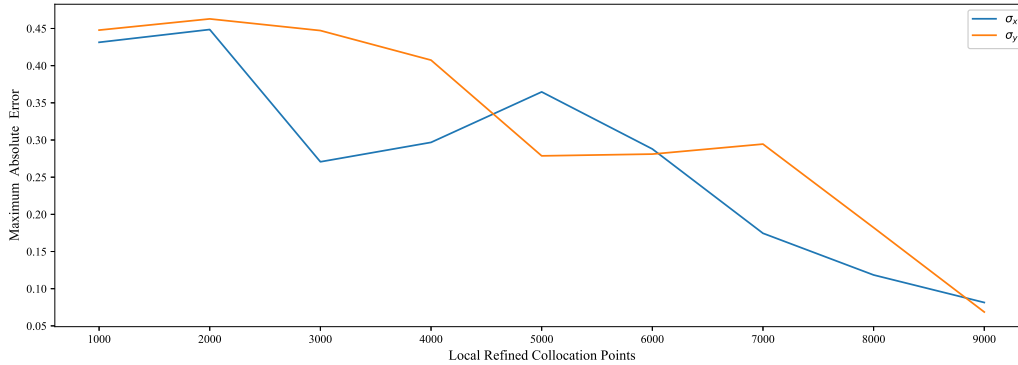


Fig. 15. Local refined sampling influence on significant gradient variations region.

This makes it advantageous in fast calculation and resource-limited scenarios.

Training Process: It is evident that there are notable distinctions between the training process of PINN and PIELM. PINN is based on iterative optimization and gradient descent, with weights being optimized through the use of a backpropagation algorithm. The training process is complex and time-consuming, and may encounter convergence issues. PIELM, in contrast, employs the training method of the extreme learning machine, which randomly generates the weights of the hidden layer and directly solves the weights of the output layer using samples at a single time point. The training process is straightforward and rapid, does not entail iterative optimization, and necessitates the resolution of a single least squares problem, with rapid computational speed.

Applicable Scenarios: In terms of application scenarios, PINN and PIELM each exhibit distinct strengths. PINN demonstrates reduced error in regions characterized by significant gradient variations and is capable of more accurate representation of regions exhibiting significant variations. However, it exhibits a relatively high overall error, thereby conferring an advantage in addressing problems of high local complexity. In contrast, PIELM demonstrates superior overall error control and is capable of providing a more precise solution on a global scale. However, it exhibits a higher error rate in regions characterized by significant gradient variations, rendering it more suited to tasks that necessitate global accuracy or applications in regions with gentle gradients.

6. Conclusions

This paper proposes and verifies a PIELM framework for solving linear elasticity problems. The proposed framework incorporates physical information into the loss function through discretization operators at collocation points, which avoids the conventional mesh generation process. The solutions are derived through least-squares approach, which is computational efficiency. Through case studies, PIELM has been demonstrated to effectively capture the physical information in the face of complex geometries and various boundary conditions. Compared to PINN, PIELM exhibits distinct advantages over PINN in analysis of plate under nonlinear and geometric defect in terms of overall error control and prediction accuracy. Compared to FME, PIELM demonstrate more precise in evaluated error norms in analysis of a three-dimensional self-weight elongation. Additionally, the findings indicate that enhancing the number of neurons in the hidden layer and local refined sampling can markedly enhance the overall and local prediction accuracy of the model.

Future research will focus on extending the PIELM framework to handle nonlinear elasticity problems, enabling it to model complex material behaviors under large deformations. Another focus will be improving solution accuracy through adaptive refinement techniques, allowing the model to dynamically adjust resolution based on problem complexity. Finally, the application of PIELM in inverse problems and parameter identification will be explored to leverage its potential for accurately estimating material properties and unknown parameters in elasticity mechanics.

CRedit authorship contribution statement

Qimin Wang: Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Chao Li:** Supervision, Resources, Formal analysis, Conceptualization. **Sheng Zhang:** Supervision, Project administration, Investigation. **Chen Zhou:** Validation, Methodology, Investigation. **Yanping Zhou:** Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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